

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');

if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(NG_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off110NGmaxFF'

```

```

selectCuts =
'Ei400_off110NGmaxFF'

```

```

if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off110NGmaxFF

```

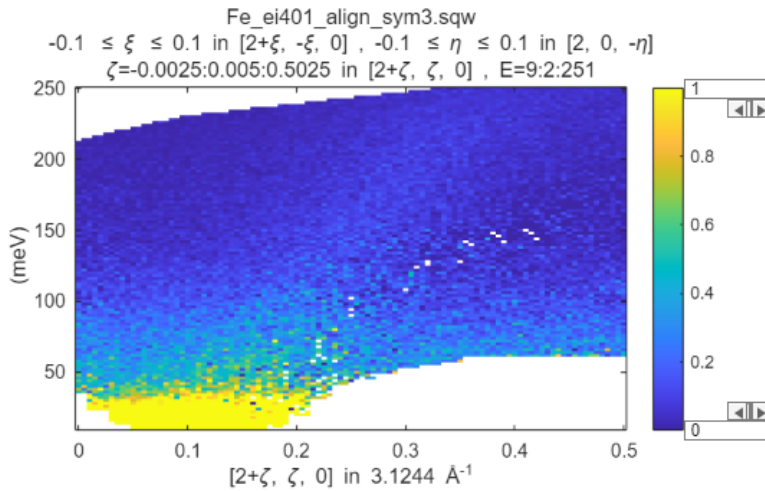
```

%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

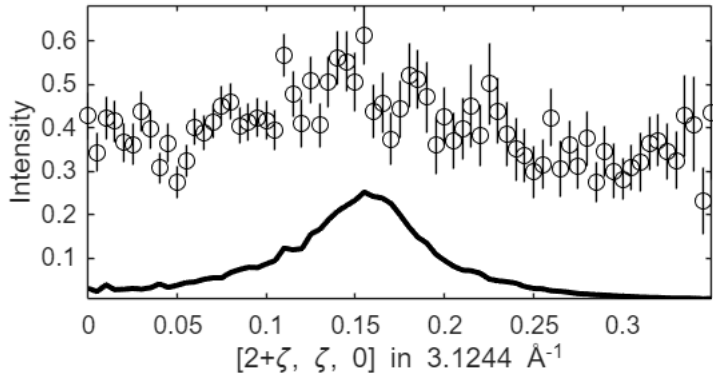
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(50,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

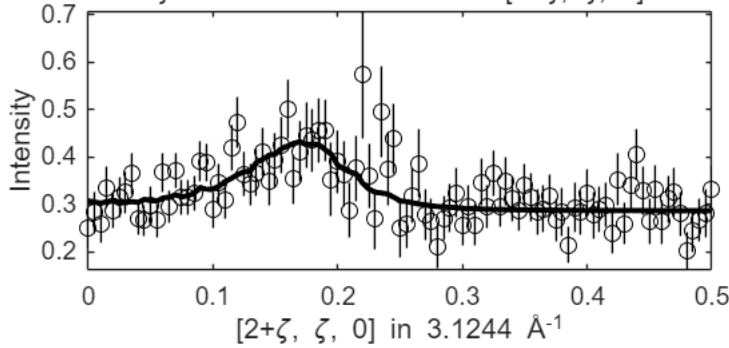
```
*****
**** En = 55±5 going to 245
*****
```

Fe_ei401_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 50
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

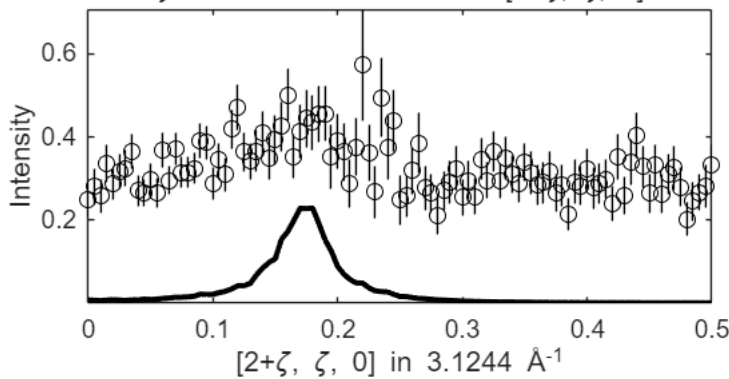


 **** En = 65±5 going to 245

Fe_ei401_align_sym3.sqw
 1.69 ± 0.066 ; $J_0 = 25 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 11.4082 \pm 1.1$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 60
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



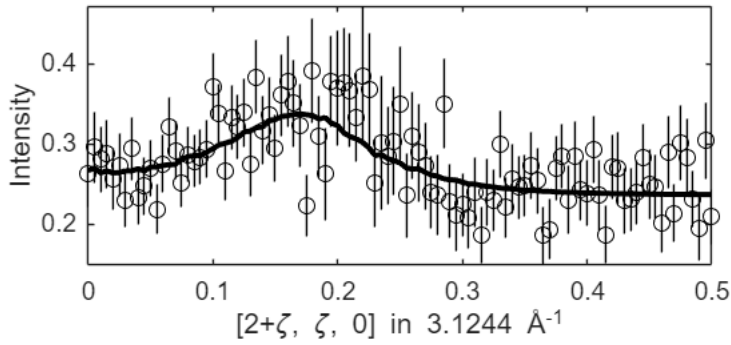
Fe_ei401_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 60
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



 **** En = 75±5 going to 245

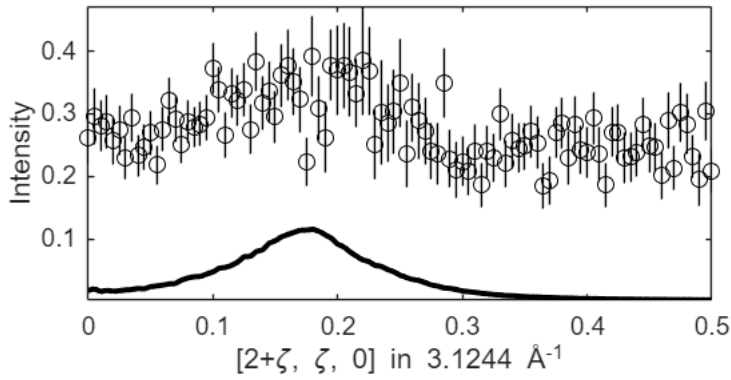
Fe_ei401_align_sym3.sqw

1.2±0.65; J0= 33± 5.8; J1= 0± 0; gamma=52.5682±41.
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 70
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

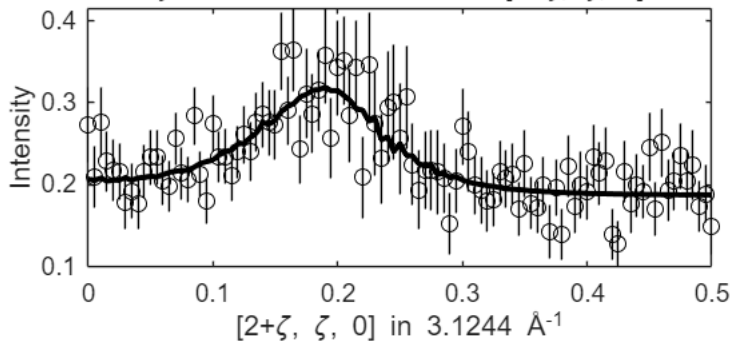
$\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 70
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



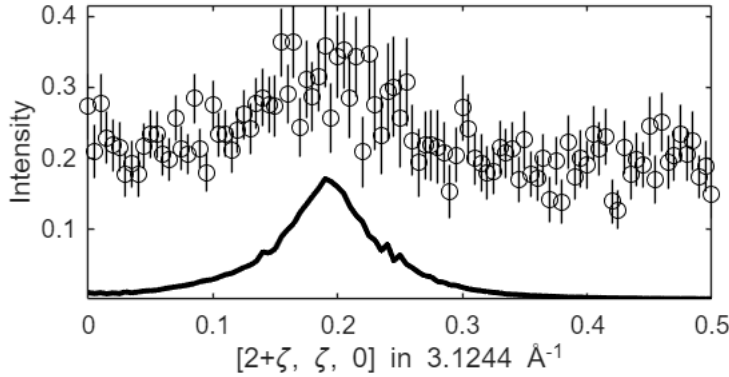
 **** En = 85±5 going to 245

Fe_ei401_align_sym3.sqw

1.1±0.076; J0= 29± 1.2; J1= 0± 0; gamma=28.1197±1.1
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 80
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

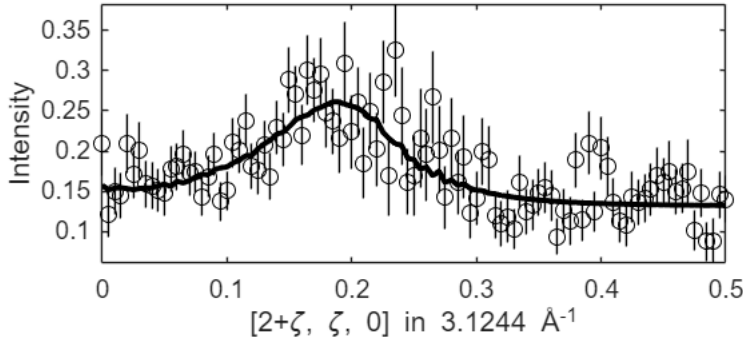


Fe_ei401_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $80 \leq \zeta \leq 90$ in $[2+\zeta, \zeta, 0]$

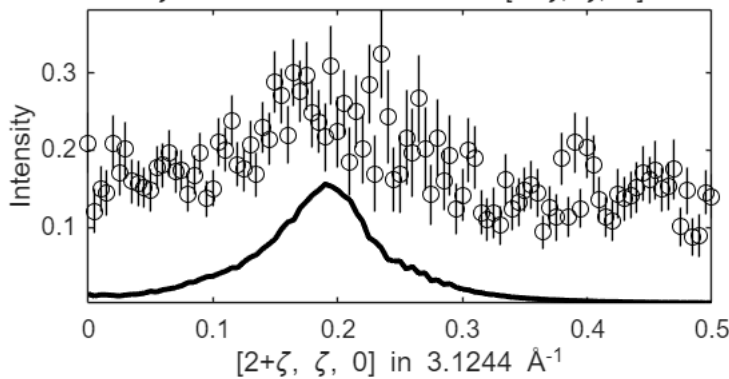


**** En = 95±5 going to 245

Fe_ei401_align_sym3.sqw
 1.3 ± 0.096 ; $J_0 = 33 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 38.8163 \pm 1.1$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $90 \leq \zeta \leq 95$ in $[2+\zeta, \zeta, 0]$



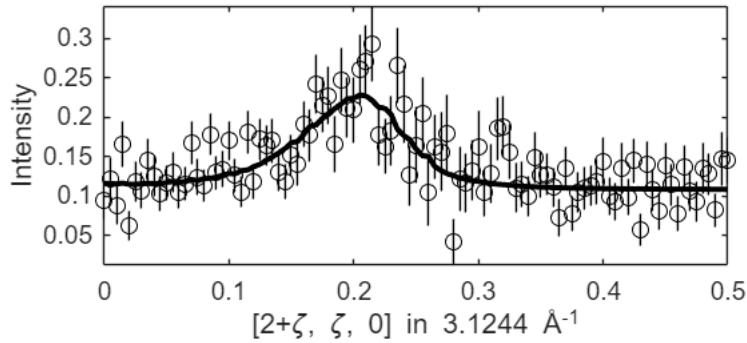
Fe_ei401_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $90 \leq \zeta \leq 105$ in $[2+\zeta, \zeta, 0]$



**** En = 105±5 going to 245

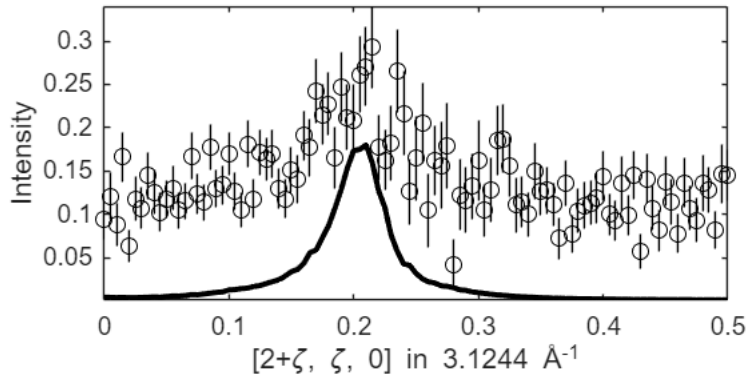
Fe_ei401_align_sym3.sqw

1.81 ± 0.072 ; $J_0 = 32 \pm 1.4$; $J_1 = 0 \pm 0$; $\gamma = 16.8578 \pm 1.0$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 100
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

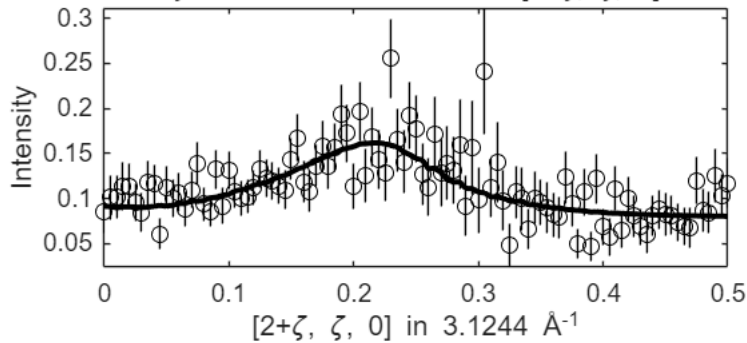
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 100
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



 **** En = 115 ± 5 going to 245

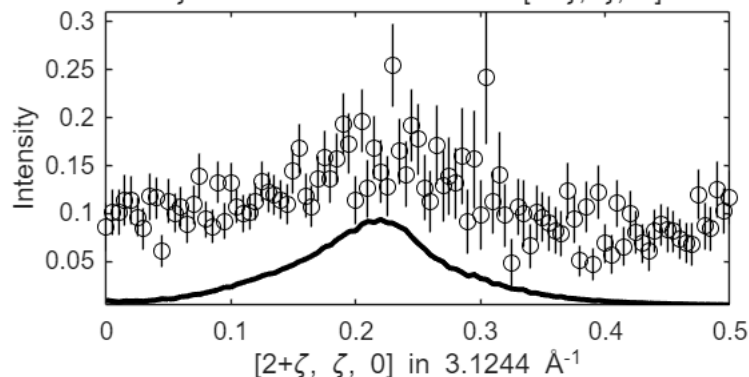
Fe_ei401_align_sym3.sqw

1.2 ± 0.11 ; $J_0 = 35 \pm 1.6$; $J_1 = 0 \pm 0$; $\gamma = 61.4909 \pm 2.0$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 110
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

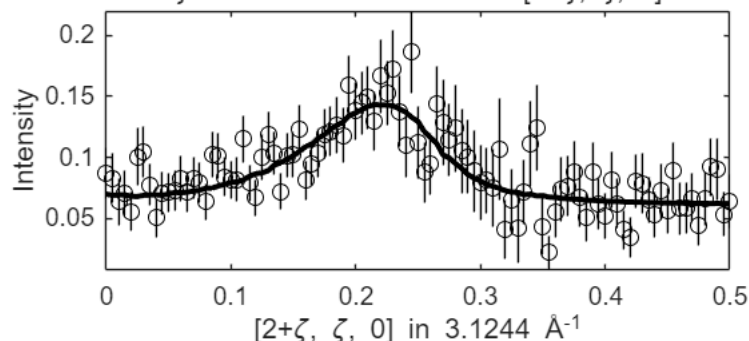
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 110 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



**** En = 125±5 going to 245

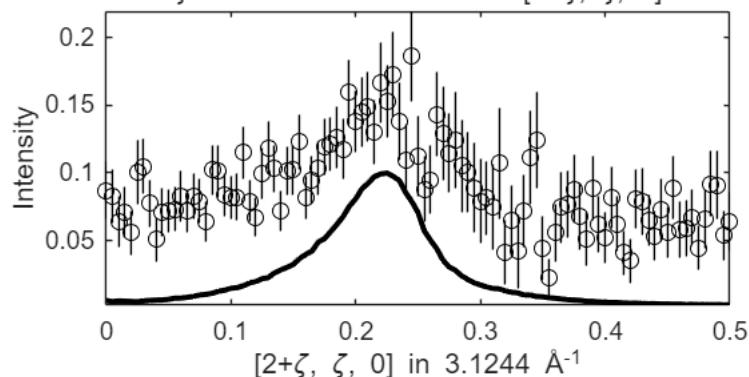
Fe_ei401_align_sym3.sqw

0.89 ± 0.2 ; $J_0 = 36 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 38.3126 \pm 13$.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 120 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

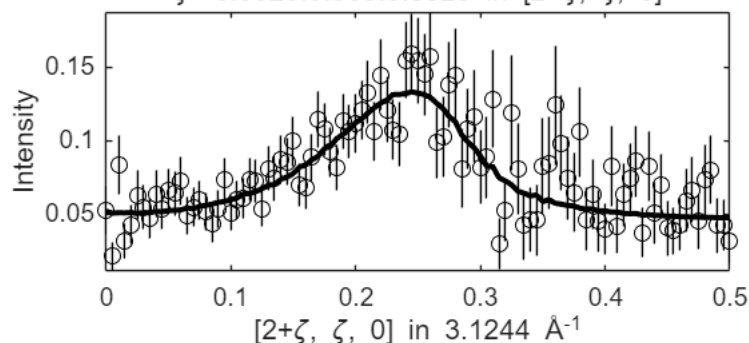
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 120 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



**** En = 135±5 going to 245

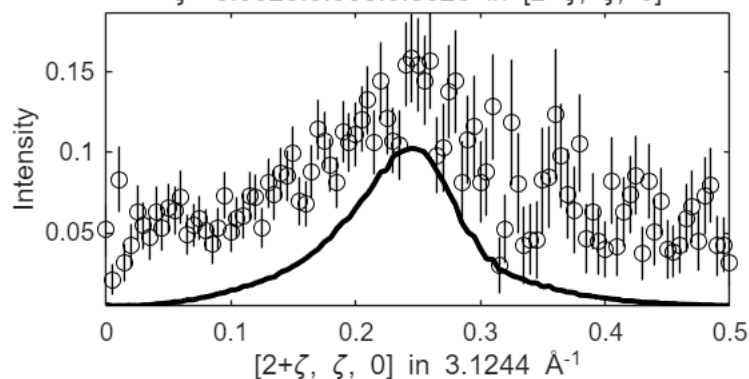
Fe_ei401_align_sym3.sqw

1.1±0.072; J0= 34±0.94; J1= 0± 0; gamma=45.6119±1.1
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 130
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 130
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

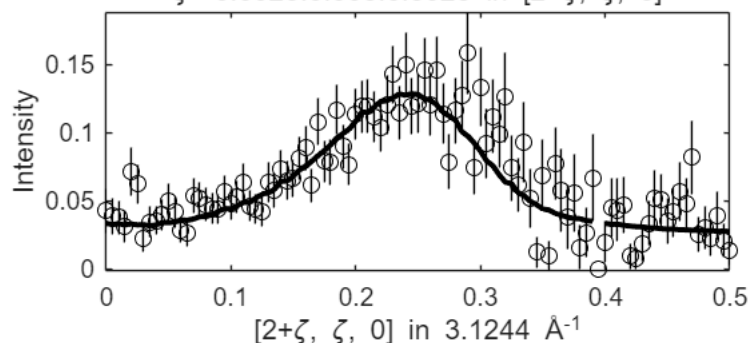


**** En = 145±5 going to 245

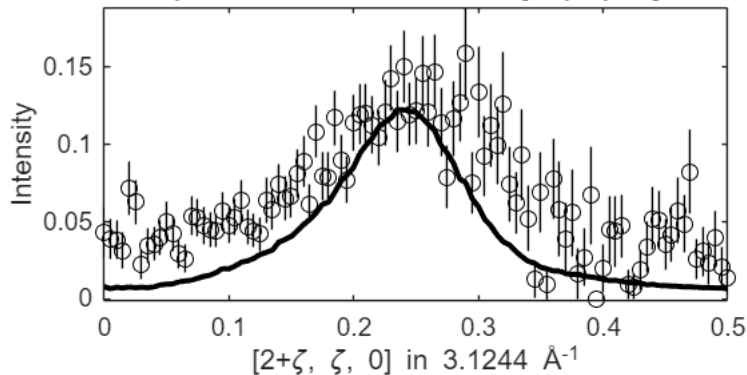
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Fe_ei401_align_sym3.sqw

1.6±0.28; J0= 37± 1.1; J1= 0± 0; gamma=56.5435±12.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 140
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

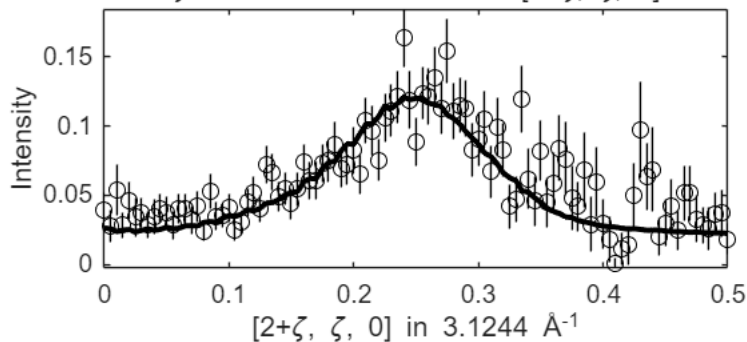


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 140 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

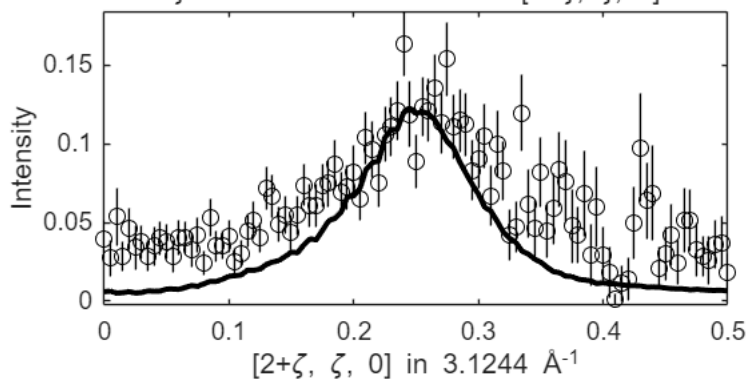


**** En = 155±5 going to 245

Fe_ei401_align_sym3.sqw
 1.5 ± 0.19 ; $J_0 = 37 \pm 0.82$; $J_1 = 0 \pm 0$; $\gamma = 49.9965 \pm 6.7$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 150 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



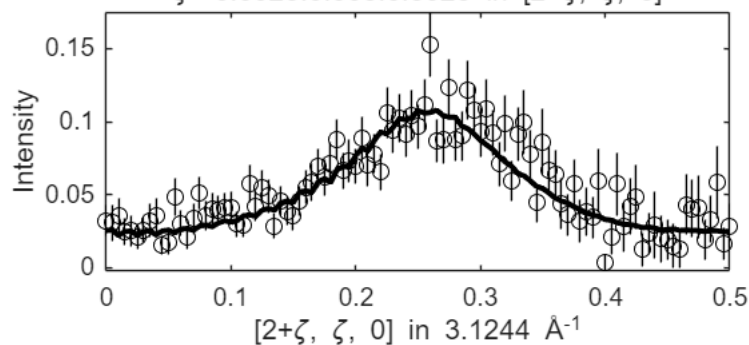
Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 150 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



**** En = 165±5 going to 245

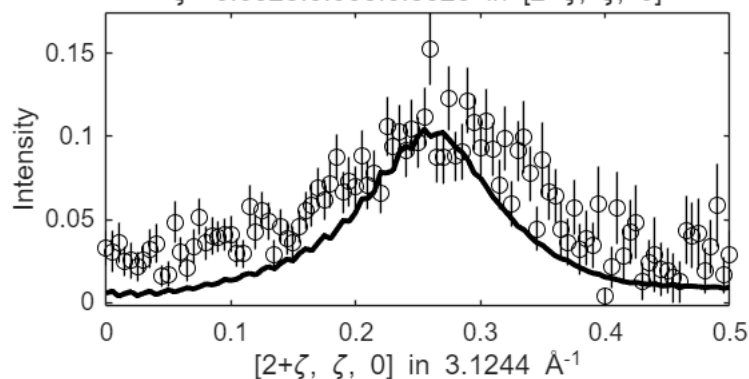
Fe_ei401_align_sym3.sqw

1.5±0.22; J0= 37±0.88; J1= 0± 0; gamma=59.3118±11.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 160
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

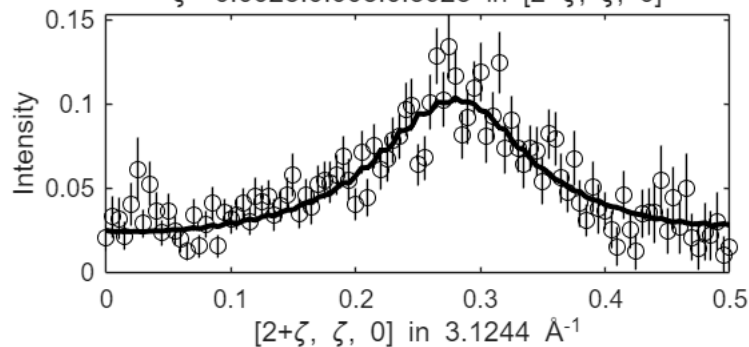
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 160
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



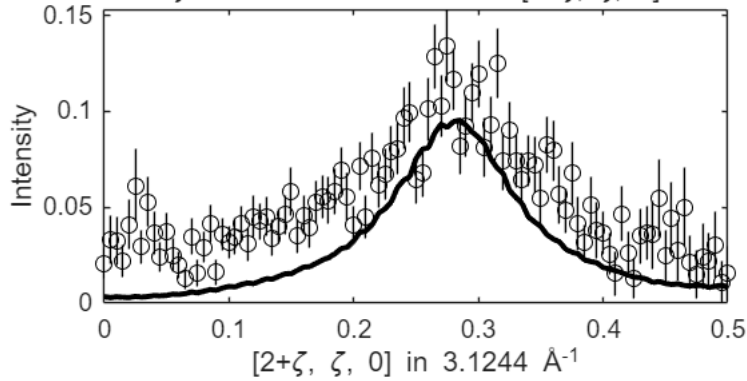
**** En = 175±5 going to 245

Fe_ei401_align_sym3.sqw

1.2±0.15; J0= 35±0.63; J1= 0± 0; gamma=48.2673±7.8
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 170
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

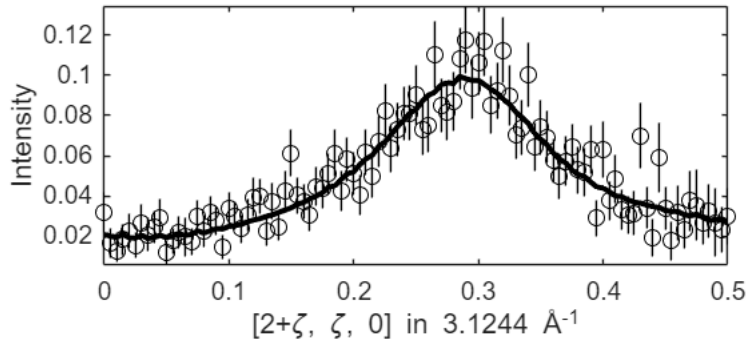


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 170
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

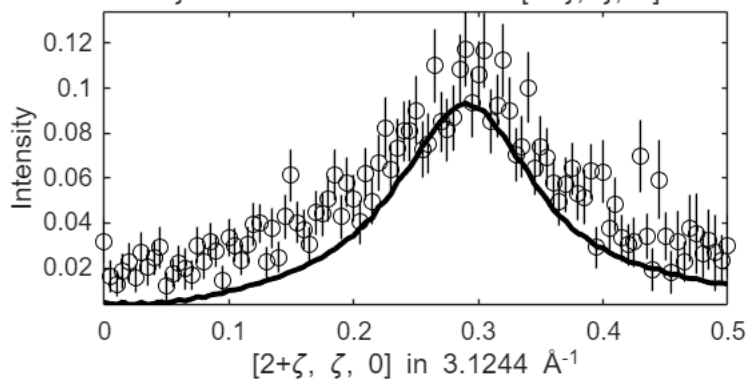


 **** En = 185±5 going to 245

Fe_ei401_align_sym3.sqw
 $= 1.4 \pm 0.13$; $J_0 = 36 \pm 0.51$; $J_1 = 0 \pm 0$; $\gamma = 60.0145 \pm 6$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 180
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



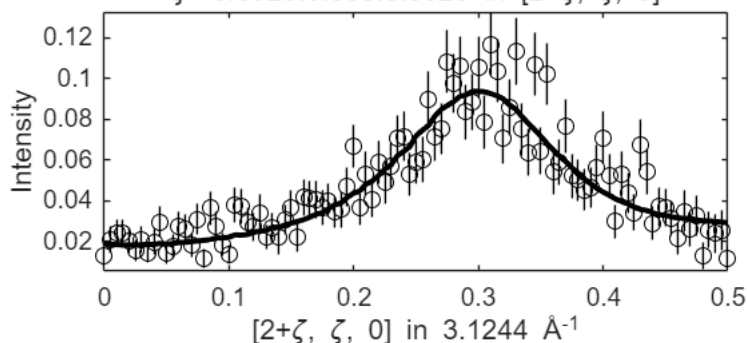
Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 180
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



 **** En = 195±5 going to 245

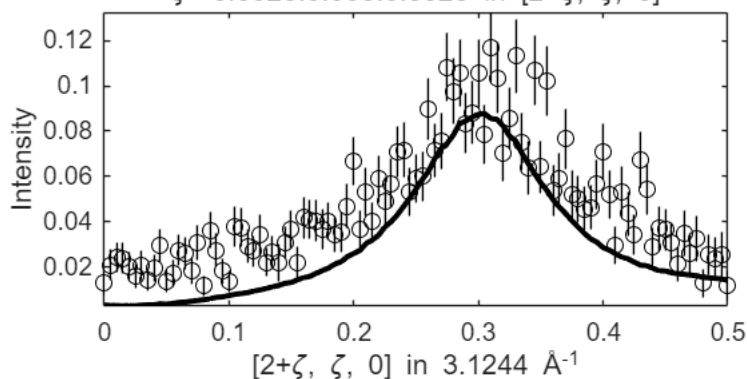
Fe_ei401_align_sym3.sqw

1.3±0.13; J0= 36±0.54; J1= 0± 0; gamma=54.8477±7.0
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 190
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

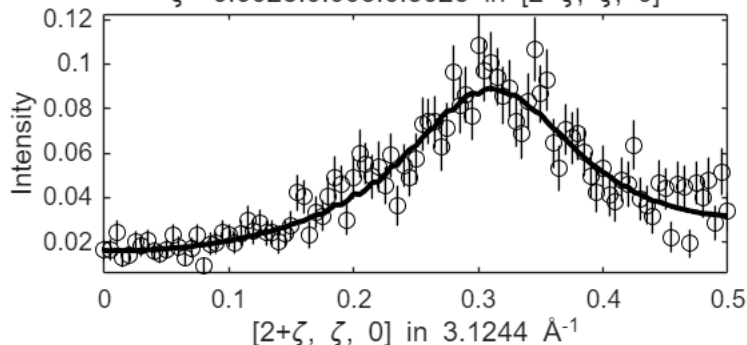
$\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 190
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



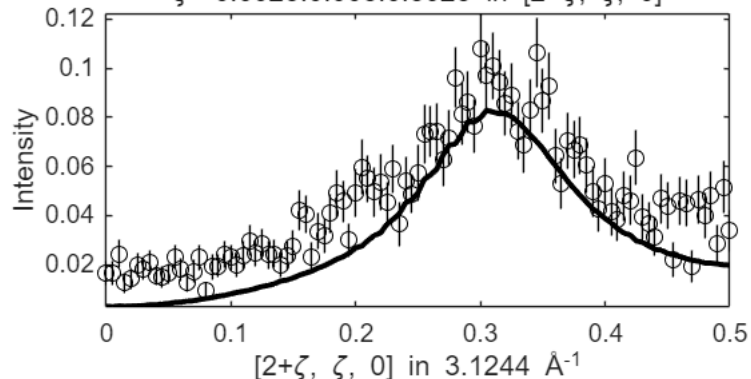
**** En = 205±5 going to 245

Fe_ei401_align_sym3.sqw

1.5±0.12; J0= 36±0.48; J1= 0± 0; gamma=65.7392±6.6
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 200
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



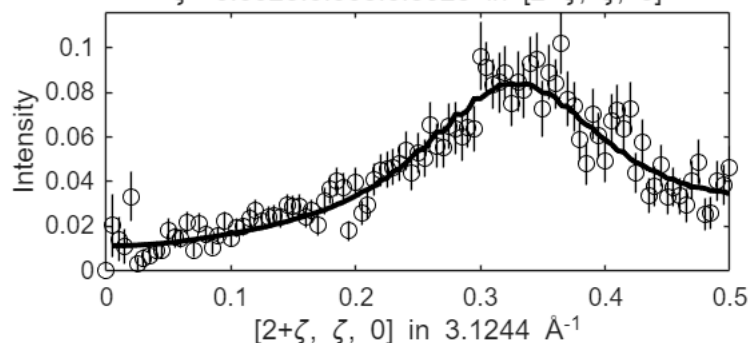
Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 200 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



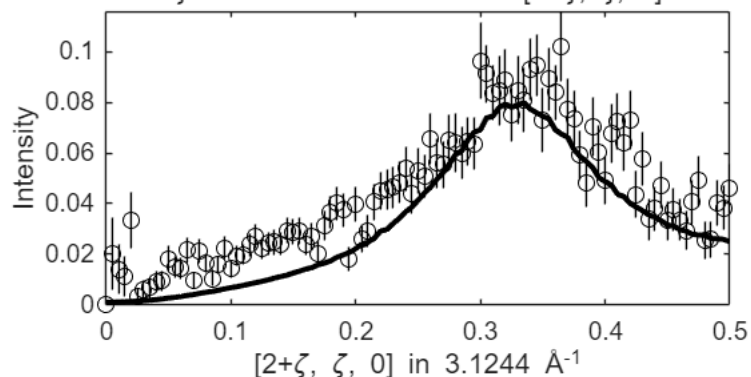
**** En = 215±5 going to 245

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Fe_ei401_align_sym3.sqw
 1.5 ± 0.097 ; $J_0 = 35 \pm 0.39$; $J_1 = 0 \pm 0$; $\gamma = 65.6961 \pm 4.$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 210 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 210 :
 $\zeta = -0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

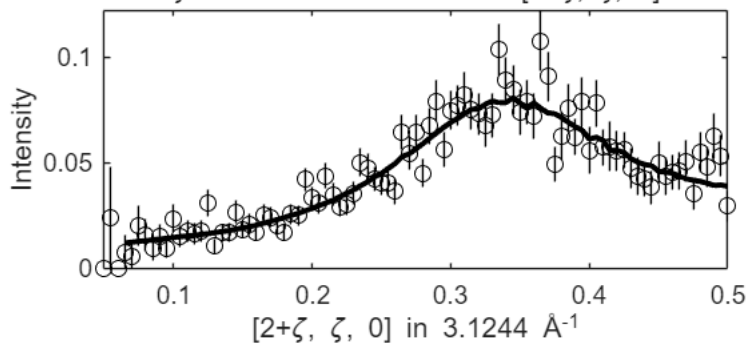


**** En = 225±5 going to 245

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802

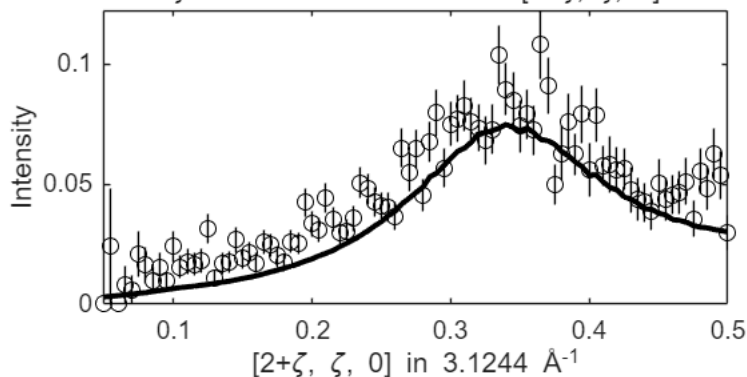
Fe_ei401_align_sym3.sqw

1.5±0.14; J0= 35±0.39; J1= 0± 0; gamma=68.1505±7.4
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 220
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 220
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

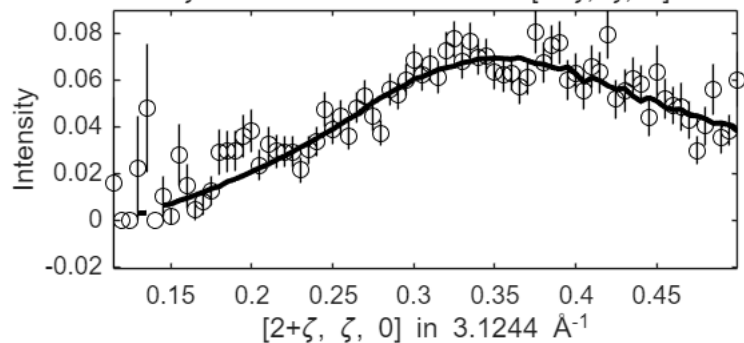


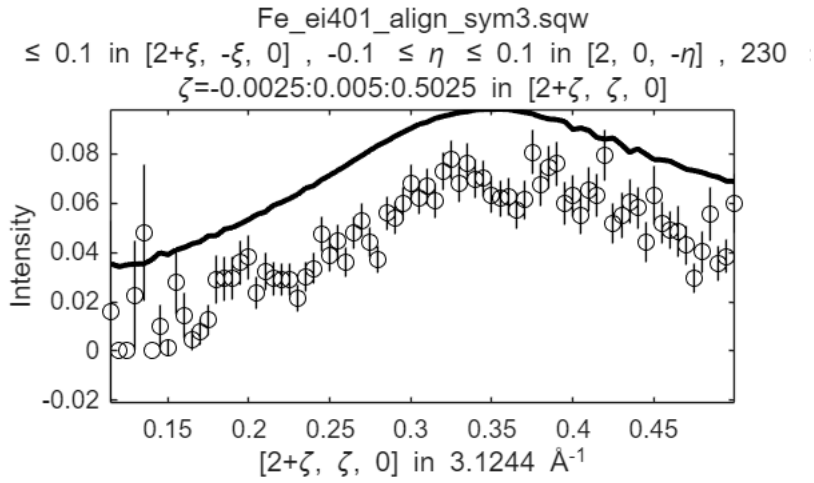
**** En = 235±5 going to 245

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.9703

Fe_ei401_align_sym3.sqw

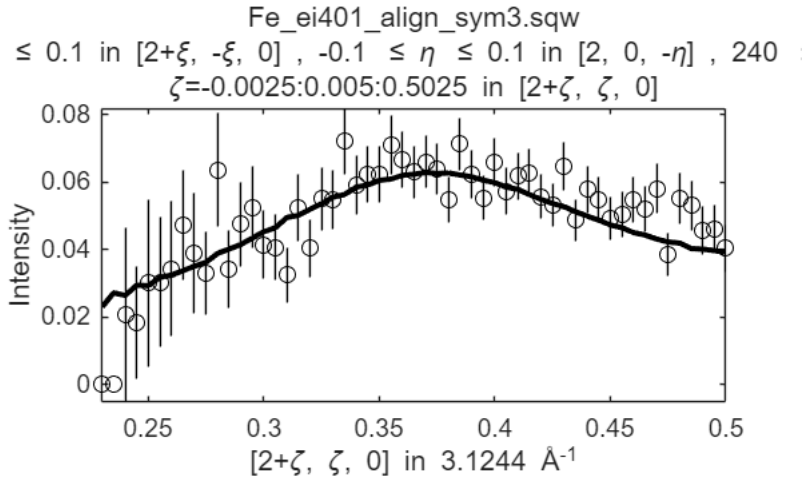
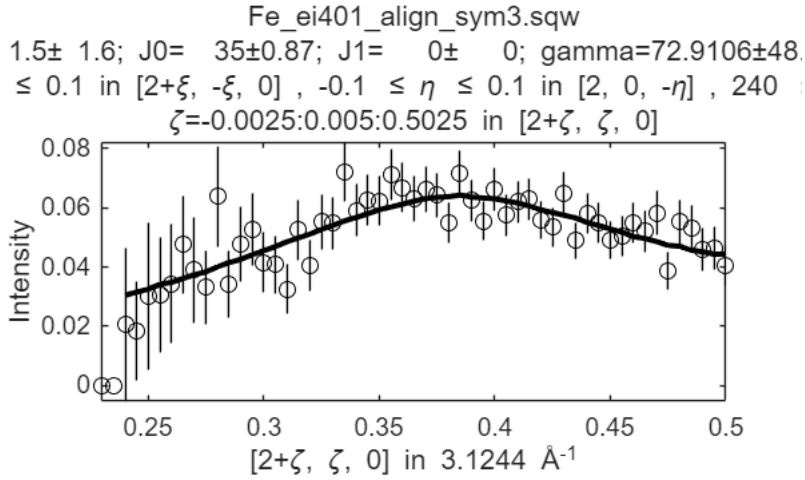
5.1±0.11; J0= 38±0.66; J1= 0± 0; gamma=188.406±10.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 230
 $\zeta=-0.0025:0.005:0.5025$ in $[2+\zeta, \zeta, 0]$

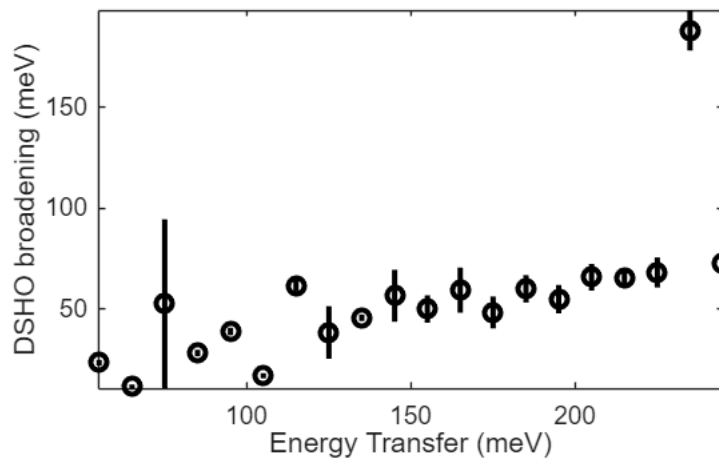
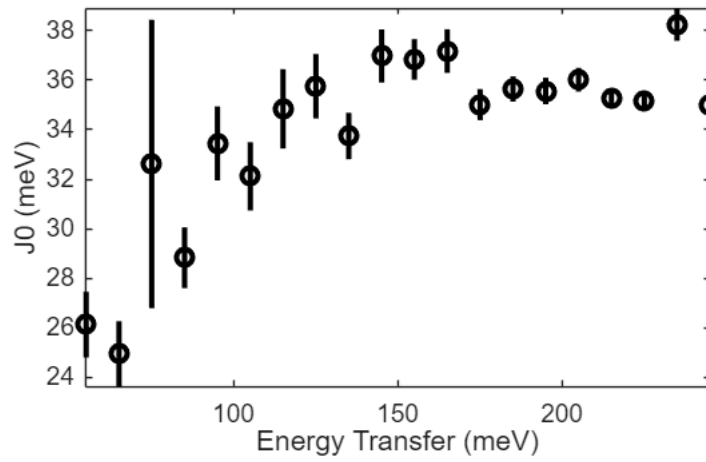
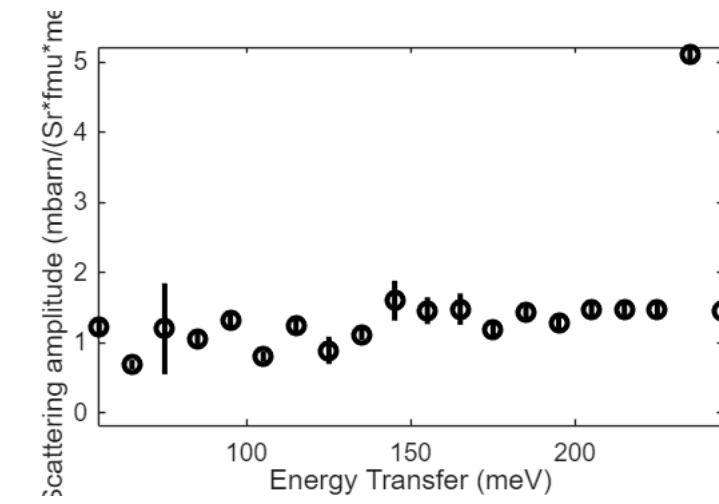




**** En = 245±5 going to 245

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802





```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

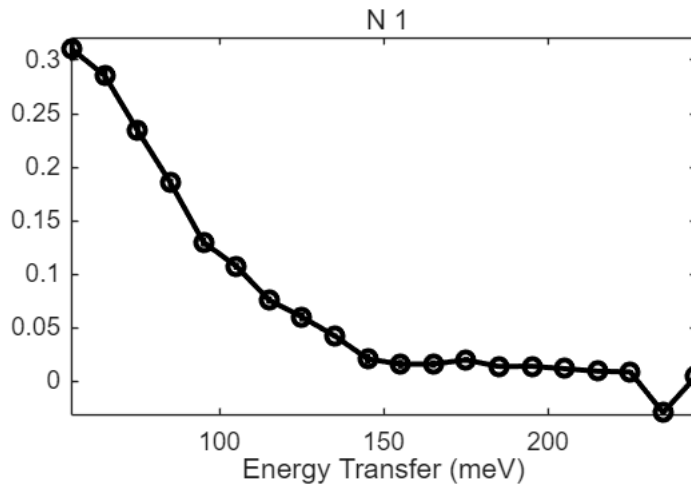
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);

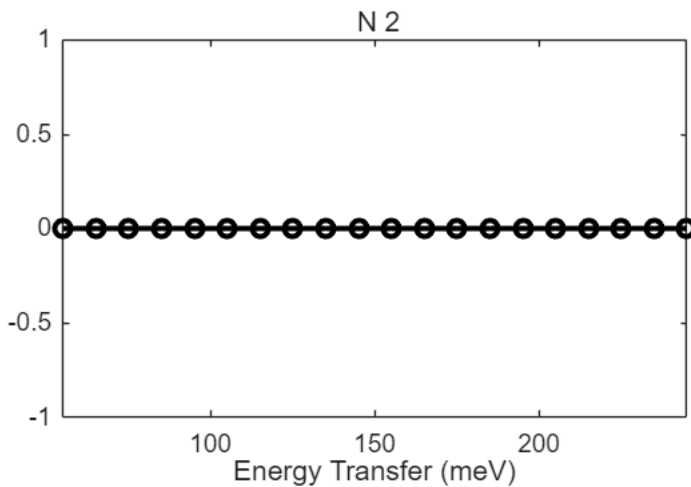
```

```
% original slope parameters
```

```
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```